

Automating Continuous Household Tidying via Mobile Manipulation

Final-Project Proposal for *Robotics*

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Abstract

We propose a *PyRoboSim*-based mobile robot that *continuously* tidies a domestic environment by fetching clutter (dirty dishes, laundry, small toys) from their spawning locations and depositing each item in its designated receptacle. A script-generated PDDL domain captures the pick–navigate–place loop, enabling a symbolic planner (Fast Downward) to produce high-level task sequences on demand. Motion is decomposed into three layers: (i) a global **RRT*** path computed with OMPL to connect semantic way-points across rooms; (ii) a real-time grid-based **A*** planner that refines each leg of the route; and (iii) a reactive **Bug2** controller that guarantees escape from unforeseen blockages. Because *PyRoboSim* supplies ground-truth object coordinates, no external perception module is required, allowing us to focus on planning, control, and life-cycle integration under ROS 2.

1 Research Question & Motivation

Problem. How can a low-cost mobile robot *continuously* locate, collect, and deliver daily clutter—dirty dishes from the living room and soiled laundry from bedrooms—to their appropriate household stations without human supervision?

Relevance. Domestic tidying consumes ≈ 1 h/day in the average European home. Automating the chore improves quality of life and forms a realistic benchmark for service-robot research.

Course linkage. Table 1 shows how each of the five lab exercises underpins the proposed system.

Ex.	Technique	Role in Project
1	2-D kinematics, joint chains	Base + 2-DOF arm forward/inverse transforms
2	Sampling motion planning (RRT*)	Global navigation between rooms
3	A*, Bug2	Local obstacle avoidance and path repair
4	STRIPS / PDDL	Symbolic task sequencing (<i>collect</i> $\rightarrow$ <i>drop-off</i>)
5	PyRoboSim/ROS 2 integration	End-to-end execution and monitoring

Table 1: Pedagogical roots of the project.

2 Proposed Solution

2.1 System Architecture

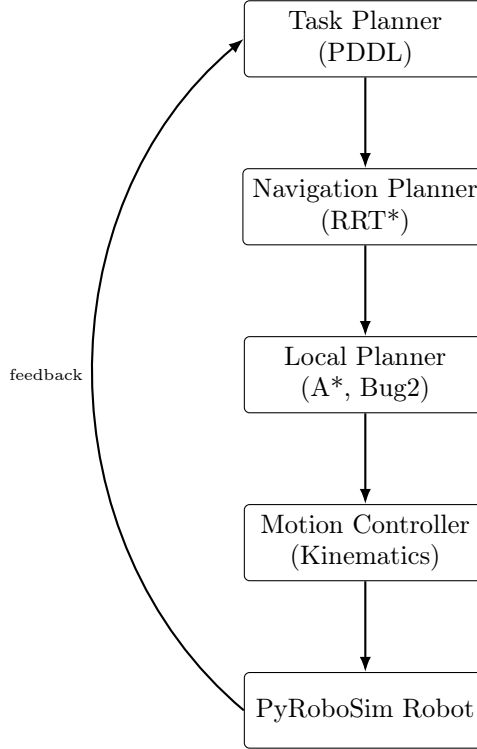


Figure 1: Layered architecture connecting symbolic planning to execution in PyRoboSim.

World model. A PyRoboSim YAML world with rooms, doorways, tables, laundry baskets, and a docking station, augmented by *temporal clutter generators* that spawn dishes/laundry via a Poisson process ($\lambda = 1/5$ hour).

Symbolic layer. PDDL domain with actions `navigate`, `pick`, `place`, `dropoff` defined in STRIPS form; plans produced by FastDownward via PyRoboSim’s planner API.

Motion layer. • **Global:** RRT* between semantic way-points.

- **Local:** A* on a 0.05 m occupancy grid; Bug2 fallback whenever dynamic obstacles break heuristic guarantees.

Execution monitor. ROS 2 action executor.

Continuity. A lifecycle node that puts the robot in service every hour.

2.2 Algorithm–Tool Mapping

Function	Algorithm / Library	Ex.
Forward kinematics	Custom Python matrices	1
Global navigation	RRT	2
Local avoidance	Bug2/A*	3
Task planning	PDDL, STRIPS	4
Simulation	PyRoboSim	5

3 Probabilistic Map-Update Example

We illustrate the belief-map update mechanism with a concise two-frame scenario (Figures 2 and 3).

Frame1 — Initial map. Figure2 depicts the known world annotated with the robot’s *a priori* probability distribution for encountering target objects. Cells shaded **red** are searched first, **orange** cells next, and **green** cells last.

Frame2 — Experience-driven update. After one day of operation the robot detects more than ten relevant objects inside a low-priority (green) region. The cell is therefore promoted to **orange** (Figure3). Analogously, any **orange** cell that accumulates 100+ detections is escalated to **red**. This threshold-based adaptation lets the system continuously realign its search strategy with real-world statistics.

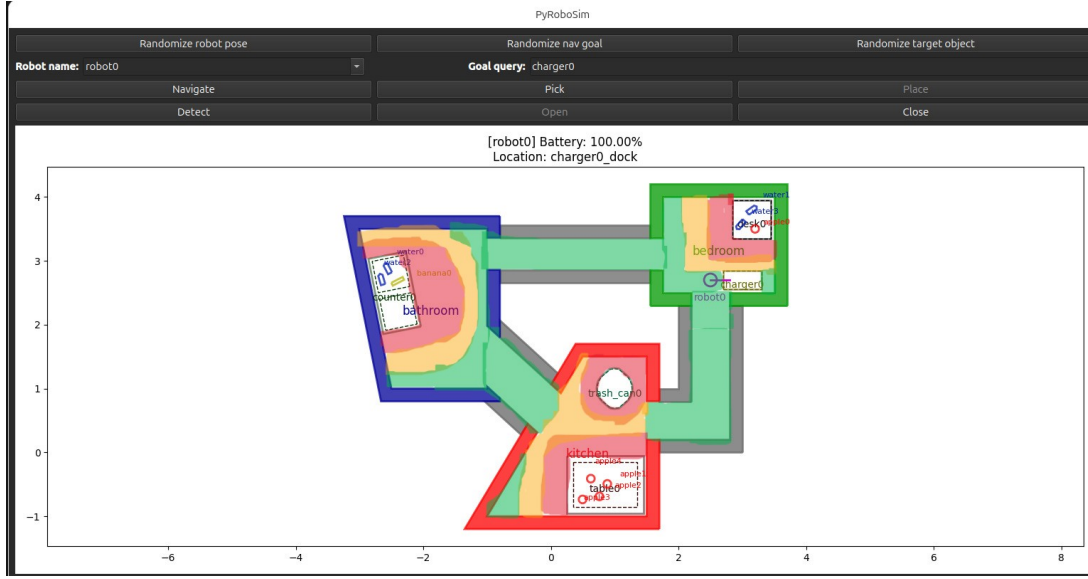


Figure 2: Initial probability map with **red** = high, **orange** = medium, **green** = low search priority.



Figure 3: Map after daily update: a previously low-priority region has been promoted to **orange** after 10+ object detections. Regions exceeding 100 detections become **red**.

4 Success Criteria & Evaluation

Metric	Target	Measurement
Task completion	$\geq 90\%$ items delivered in a service round	Logged world-state queries
Planning time	Symbolic $< 10s$; global path $< 5s$	ROS 2 timestamps
Path optimality	Less then predefined max time	Compare to heuristic bound
Git hygiene	$\geq 90\%$ unit-test coverage; green CI	GitHub Actions

A run is **successful** if all four metrics meet their targets.

5 Project Plan & Management

5.1 Timeline

Week	Milestone	Deliverables
1	World-building	YAML house, clutter script
2	Global planner	RRT* node, benchmark notebook
3	Local planner	A* grid & Bug2, collision tests
4	PDDL domain	STRIPS operators, unit plans
5	Integration	End-to-end demo video
6	Evaluation	Metric harness, stress tests
7	Report & polish	Final paper, code freeze

5.2 Tools & Practices

- **Version control:** GitLab, protected `main`, squash PRs.
- **CI:** GitLabActions—lint, pytest, headless simulation, coverage.
- **Issue tracking:** GitLab Projects kanban, label taxonomy.
- **Docs:** MkDocs auto-deployed to GitLab Pages.
- **Risk mitigation:** weekly 20-min review of codes.

6 References

References

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